

23/09/

Experiment No:-4

NOVOTEL

HYDERABAD CONVENTION CENTRE

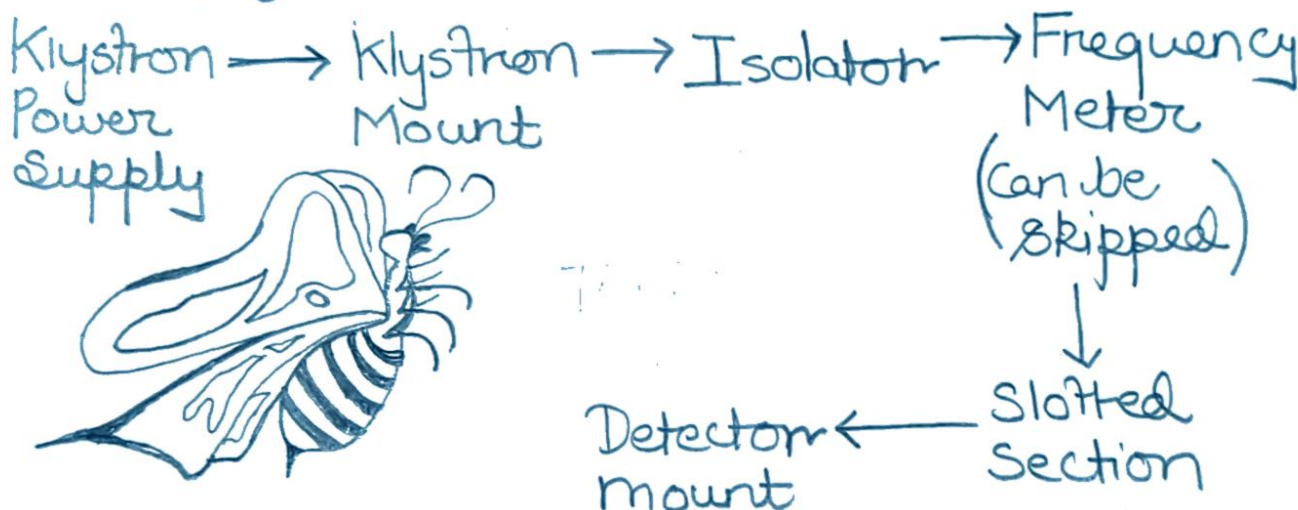


HYDERABAD INTERNATIONAL
CONVENTION CENTRE

Managed by A.

Objective:- Measurement of
dielectric constant of a
given material (solid).

Block Diagram:-



*Initial Klystron Power Supply

1. Beam Voltage : Minimum ; 260 Volts
2. Repeller Voltage : Maximum ; Drop it to attain 1st mode.
3. Current : 10 milliamp
4. Frequency : 9.595 GHz
5. Modulation : AM
6. Dielectric Sample : PDMS (Polydimethylsiloxane)
7. Dimension of Wave guide :- $L \times W :- 1 \times 2.3 \text{ cm}$
8. Dimension of Sample :- $L \times W \times H ; H = 0.6 \text{ cm}$

GUEST USE STATIONERY

*Calculation

» First minima without dielectric (d_1) = (MSR)

» Second minima without dielectric (d_2) = (MSR)

» Guided Wavelength (λ_g) = $2(d_2 - d_1)$ =

» Change in first minima with = dielectric sample (d_3)

» Δs , difference between first minima with or without dielectric sample

$$\Rightarrow \Delta s = d_3 - d_1$$

$$\Rightarrow X'' = \left(\frac{\lambda_g}{H} \right) \tan(2\pi(\Delta s + H)/\lambda_g)$$

$$\Rightarrow \left(\frac{1}{\lambda_0} \right)^2 = \left(\frac{1}{\lambda_g} \right)^2 + \left(\frac{1}{\lambda_c} \right)^2 \quad \left\{ \begin{array}{l} \text{Determining the} \\ \text{frequency} \end{array} \right\}$$

» Make use of Graph

$$\Rightarrow X'' = \frac{\tan(X)}{X}$$

$$\Rightarrow X = 2\pi V$$

$$\gg \text{Dielectric Constant } (\epsilon_r) = 1 - \left(\frac{\lambda_0}{\lambda_g} \right)^2 + \left(\frac{\lambda_0 V}{H} \right)^2$$

Now you can compare the value with reported work on some good research paper or any reference website or text book.

» $\sigma = 0$, the medium is a perfect or lossless, dielectric.

For $\sigma \neq 0$, the medium is a lossy or imperfect, dielectric

» $\omega\epsilon \gg \sigma$, it behaves more like a dielectric than anything else, called as a dielectric.

» The conduction current is much greater than the displacement current called as conductor.

$\frac{\sigma}{\omega\epsilon} < \frac{1}{100}$; $\frac{\sigma}{\omega\epsilon}$ is dimensionless.

σ = conductivity of medium, Ωm^{-1}

ϵ = permittivity of medium, $F m^{-1}$

ω = radian frequency = $2\pi f$

f is in Hz.

» $d_1 = 8.3$ ——— X ——— X ——— X ———

$$\Rightarrow d_2 = 10.2$$

$$\Rightarrow \lambda_g = 2(10.2 - 8.3) = 3.8$$

$$\Rightarrow d_3 = 9.4$$

$$\Rightarrow \Delta S = 9.4 - 8.3 = 1.1$$

$$\Rightarrow X'' = \left(\frac{\lambda_g}{H} \right) \tan(2\pi(\Delta S + H)) / \lambda_g$$

$$\Rightarrow \left(\frac{1}{\lambda_0} \right)^2 = \left(\frac{1}{\lambda_g} \right)^2 + \left(\frac{1}{\lambda_c} \right)^2$$

$$f = \frac{c}{\lambda_0} = 10.2 \text{ GHz}$$

$$\lambda_c = 2 \times 2.3 = 4.6$$

$$\Rightarrow \left(\frac{1}{\lambda_0} \right)^2 = \left(\frac{1}{3.8} \right)^2 + \left(\frac{1}{4.6} \right)^2$$

$$\Rightarrow \frac{1}{\lambda_0^2} = \frac{1}{14.44} + \frac{1}{21.16}$$

$$\Rightarrow \sqrt{\frac{1}{0.1165110568}} = \lambda_0 = 2.92 \text{ cm} = 1.02$$

$$\Rightarrow X'' = \left(\frac{3.8}{0.6} \right) \tan \left(\frac{2\pi(1.1 + 0.6)}{3.8} \right)$$

$$\tan(2.81089869)$$

$$= \left(\frac{3.8}{0.6} \right) (0.04909883422)$$

$$\Rightarrow X'' = 0.310; \frac{\tan(X)}{X} = X'' \Rightarrow X = 2.6$$

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• Principals: CST Studio Suite - 3D Electromagnetic Simulation Software, Mechanical, Structural, Material Analysis Software, RF Connectors, Cable Assemblies, Amplifiers, Active and Passive Components, Venable-Frequency Response Analyzer, IP Solutions-NavIC Receiver and Simulator, Winton Cable Bending Machine, Rohde & Schwarz VNA & Measurement Instruments

Dielectric Constant

$$\Rightarrow \epsilon_r = 1 - \left(\frac{\lambda_0}{\lambda_g} \right)^2 + \left(\frac{\lambda_0 V}{H} \right)^2$$

$$\Rightarrow x = 2\pi V$$

$$\Rightarrow \frac{x}{2\pi} = V = 0.6525352667 \approx 0.65$$

$$\Rightarrow \epsilon_r = 1 - \left(\frac{2.92}{3.8} \right)^2 + \left(\frac{2.92 \times 0.65}{0.6} \right)^2$$

$$\Rightarrow \epsilon_r = 4.3; \text{ Polydimethylsiloxane (PDMS).}$$

Actual Value ≈ 2.3 !

—X—X—X—